

SPECIAL ARTICLE

Society of Critical Care Medicine Guidelines for the Allocation of Critical Care Resources to Adults During Crisis-Level Shortages

RATIONALE: Efficient distribution of scarce critical care resources is essential to save the most lives in times of crisis. Evidence-based practices and processes enhance clinical decision-making.

OBJECTIVES: The objective of these guidelines was to develop evidence-based, rather than expert-based, recommendations for triaging critically ill patients eligible for ICU admission during times of crisis-level shortages in ICU capacity.

DESIGN: The American College of Critical Care Medicine Board convened a 21-member multidisciplinary panel, comprising doctors in medicine, nursing, and law; advanced practice providers; respiratory therapists; ethicists; and patient/family representatives. The panel included two expert methodologists specialized in developing evidence-based recommendations in alignment with the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) methodology. Conflict-of-interest policies were strictly followed during all phases of guidelines development including task force selection and voting.

METHODS: The panel members identified and formulated five fundamental Patient, Intervention, Comparator, and Outcomes questions. The panel conducted a systematic review for each question to identify the best available evidence, analyzed the evidence, and assessed the certainty of the evidence using the GRADE methodology. The GRADE evidence-to-decision framework was used to formulate the recommendations. Good practice statements were included to provide additional guidance.

RESULTS: The panel generated one conditional recommendation and five no recommendation statements.

CONCLUSIONS: Crisis-level shortages significantly disrupt patient care. Despite the role of triage in minimizing adverse outcomes, there is a lack of evidence, as opposed to expert opinion, to guide practice recommendations in the critical clinical scenarios considered by the panel.

KEYWORDS: adults; clinical practice guidelines; intensive care unit; resource allocation; surge capacity; triage

Triage is the complex process of prioritizing the allocation of critical resources, and it is essential during situations involving crisis-level shortages in the ICU (1). Crisis-level shortages refer to severe deficits of essential resources and services—including products (e.g., blood, ventilators), personnel (e.g., respiratory therapists, nurses), and facilities (e.g., ICU beds)—that drastically affect the delivery of care to critically ill patients. ICU capacity is the maximum number of patients that an ICU can accommodate while maintaining a high level of specialized care and monitoring. The capacity of an ICU depends on multiple factors, such as physical space, availability of

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DOI: 10.1097/CCM.0000000000006998

medical equipment, staffing levels, and the expertise of healthcare professionals. Prioritizing patients also requires managing these resources (e.g., patient-flow and other system inefficiencies) through a process called flow-sizing, or matching capacity and demand to ensure that all patients receive appropriate care (2, 3).

In 1994, the Society of Critical Care Medicine (SCCM) Ethics Committee established the basic principles of triage (e.g., meet ICU admission criteria, expected survival benefit) (4). These principles have remained valid to this day. In 2016, a SCCM taskforce updated the 1999 ICU admission and triage guidelines (5) and made extensive recommendations (6). In an effort to continuously improve the triage process, the updating of these guidelines was undertaken. Other key definitions are included in **Table S1** (<https://links.lww.com/CCM/H862>).

METHODOLOGY

In 2023, the American College of Critical Care Medicine, the consultative body of SCCM, commissioned a panel of experts in critical care medicine to develop focused, evidence-based recommendations for triaging critically ill patients eligible for ICU admission during times of crisis-level shortages in ICU capacity. This taskforce included physicians, advanced practice providers, nurses, respiratory therapists, an ethicist, and other allied health and/or administrative professionals. Additionally, two methodologists from the Guidelines in Intensive Care Development and Evaluation Group of McMaster University supported guidelines development. Members of the panel were required to disclose conflicts of interest (COI) per the SCCM's COI policy. COI was assessed at each phase of the guidelines process and at every panel meeting. Panel members with a relevant COI were excluded from voting on the recommendation.

The panel conducted a systematic review of the published scientific literature, focusing on patient-oriented, clinically relevant outcomes to answer Patient, Intervention, Comparator, and Outcomes (PICO) questions regarding the triage of critically ill adults in the ICU. For each PICO question, the panel identified three subgroups of interest: 1) normal operations, 2) short-term, and 3) long-term crisis-level shortages. The final PICO questions are presented in

Table 1. The panel rated the relative importance of each outcome to determine which outcomes were critical or important for decision-making. A summary of the searched outcomes for all PICO questions is presented in **Table S2** (<https://links.lww.com/CCM/H862>). The panel performed an extensive review of the scientific literature through November 2, 2023, to retrieve articles that addressed the PICO questions. The search was conducted by a medical librarian using MEDLINE, Embase, CENTRAL, ClinicalTrials.gov, and World Health Organization International Clinical Trials Registry Platform (Preferred Reporting Items for Systematic reviews and Meta-Analyses diagrams, **Figs. S1–S4**, <https://links.lww.com/CCM/H862>, are provided, as well as the search strategies and queries in the **Supplemental Materials**, <https://links.lww.com/CCM/H862>). Search results were then uploaded to Covidence for screening, full-text review, and data extraction (7).

Data extraction and risk of bias assessment was performed independently and in duplicate by two experienced methodologists using pre-piloted forms. Conflicts were resolved by discussion between methodologists until consensus was reached. The risk of bias for randomized controlled trials (RCTs) was assessed using the Cochrane Risk of Bias 2 for RCTs (8) and Clinical Advances through Research and Information Translation (CLARITY; McMaster University, Hamilton, ON, Canada, 2025) tool for observational studies (9). Meta-analyses were performed on the outcomes of interest where possible, for each PICO question using a web-based meta-analysis platform (10). Data that could not be quantitatively pooled or outcomes with insufficient studies to analyze were summarized narratively.

The clinical practice recommendations were then developed according to the Grading of Recommendations, Assessment, Development, and Evaluation process (11). The Evidence-to-Decision framework was completed by the panel using GradePro software (12) for each PICO to develop a draft recommendation considering the balance of desirable and undesirable effects, certainty of the evidence, resource considerations, feasibility, acceptability, and equity considerations (see Supplemental Materials, <https://links.lww.com/CCM/H862>, for evidence-to-decision worksheets). Recommendations had to receive at least 80% consensus agreement by at least

TABLE 1.
ICU Triage Patient, Intervention, Comparator, and Outcomes Questions

Question 1. In situations of limited ICU bed availability for critically ill adult patients, should clinician judgment for prioritization of admission be used over time-based admission to ICU?			
Population	Intervention	Comparator	Outcomes
Adult critically ill patients (age ≥ 18 yr old defined by an ICU admission request or order as per local hospital processes) Setting: Limited ICU bed availability as defined by included studies Subgroups of interest 1) Normal operational surge conditions ($> 90\%$ of ICU beds are occupied) 2) Short-lived disasters (hurricanes, etc) 3) Longer-term disasters (pandemics, etc)	Clinician judgment (defined as use of physician assessment, patient factor based, or severity of illness) system or strategy for prioritization of admissions to the ICU	Time-based admission (first-come, first-served) to ICU (patients that "meet ICU criteria" are admitted in the order of arrival or referral to ICU)	See Supplemental Materials and Table S2 (https://links.lww.com/CCM/H862)
Question 2. In situations of limited ICU bed availability for critically ill adult patients, should a formal triage tool be used vs. no triage tool?			
Population	Intervention	Comparator	Outcomes
Adult critically ill patients (age ≥ 18 yr old defined by an ICU admission request or order as per local hospital processes) Setting: Limited ICU bed availability as defined by included studies Subgroups of interest 1) Normal operational surge conditions ($> 90\%$ of ICU beds are occupied) 2) Short-lived disasters (hurricanes, etc) 3) Longer-term disasters (pandemics, etc)	Objective triage tool (e.g., Sequential Organ Failure Assessment) Intervention subgroups of interest (subject to data availability): 1) Triage system using machine learning/artificial intelligence 2) Diagnosis-based 3) Other specific triage tools	No objective triage tool	See Supplemental Materials and Table S2 (https://links.lww.com/CCM/H862)
Question 3A. In situations of limited ICU bed availability for critically ill adult patients, should patients awaiting an ICU bed be transferred to another facility?			
Question 3B. For patients not transferred to another facility, should hospitals with ICUs have a designated non-ICU area that is prepared to board ICU patients during surge conditions vs. no designated non-ICU area?			

(Continued)

TABLE 1. (Continued)
ICU Triage Patient, Intervention, Comparator, and Outcomes Questions

Population	Intervention	Comparator	Outcomes
Adult critically ill patients (age ≥ 18 yr old defined by an ICU admission request or order as per local hospital processes) Setting: Limited ICU bed availability as defined by included studies Subgroups of interest 1) Normal operational surge conditions 2) Short-lived disasters (hurricanes, etc) 3) Longer-term disasters (pandemics, etc)	Interfacility transfer of patients	No interfacility transfer	See Supplemental Materials and Table S2 (https://links.lww.com/CCM/H862)
Hospitals with ICUs Setting: Limited ICU bed availability as defined by included studies Subgroups of interest 1) Normal operational surge conditions 2) Short-lived disasters (hurricanes, etc) 3) Longer-term disasters (pandemics, etc)	Having a designated non-ICU area that is prepared in advance of surges to board ICU patients during surge conditions Potential subgroups: 1) ED (e.g., ED-ICU) 2) Designated overflow areas (e.g., post-anesthesia care unit, operating room, intermediate care unit, etc) 3) Specialty ICUs where admitting diagnosis is discordant to the specialty (e.g., chronic obstructive pulmonary disease exacerbation admitted to surgical ICU), if available.	Standard care (no designated non-ICU area)	See Supplemental Materials and Table S2 (https://links.lww.com/CCM/H862)
Question 4. In situations of limited ICU bed availability for critically ill adult patients, should patients boarded in non-ICU care areas be managed by ICU trained practitioners or by the usual practitioners for that area (i.e., emergency medicine physician, anesthesiologist, hospitalist)?			
Population	Intervention	Comparator	Outcomes
Adult critically ill patients (age ≥ 18 yr old defined by an ICU admission request or order as per local hospital processes) Setting: Patients boarded outside ICU	Managed by usual ICU practitioners	Managed by non-ICU practitioners (e.g., ED, anesthesiologist, hospitalists, etc) Non-ICU practitioners supervised by ICU practitioners	See Supplemental Materials and Table S2 (https://links.lww.com/CCM/H862)
Question 5. During situations of limited ICU bed availability, in critically ill patients at high risk of dying in the ICU, should palliative care services be involved early vs. at the usual clinician discretion?			
Population	Intervention	Comparator	Outcomes
Critically ill patients at high risk of dying in the ICU during situations of limited ICU capacity Setting: Limited ICU bed availability as defined by included studies	Early involvement of palliative care service	Routine or no involvement of palliative care service	See Supplemental Materials and Table S2 (https://links.lww.com/CCM/H862)

ED = emergency department.

75% of the panel to be approved. These clinical practice guidelines reflect the state of knowledge at the time of publication.

RECOMMENDATIONS

Each recommendation statement was assigned a strength (“Strong” or “Conditional”). A “Strong” recommendation is one that clinicians should follow for nearly all patients (i.e., something that might qualify as a quality measure). A “Conditional” recommendation reflects a lower degree of certainty in the appropriateness of the patient care strategy. It requires that the clinician use clinical knowledge and expertise and strongly considers the individual patient’s values and preferences to determine the best course of action.

PICO 1. Clinician Judgment Vs. Time-Based Admission

We make no recommendation regarding the use of clinician judgment over time-based prioritization for ICU admission in situations of limited ICU bed availability due to insufficient evidence.

Rationale. During surge conditions, determining an effective ICU admission prioritization method is essential to optimize patient outcomes. Clinician judgment incorporating all basic components of triage (13, 14), the time-based first come, first served or a combination of these two methods are the most often used strategies. Understanding who benefits from ICU admission, which would further inform clinician judgment-based approaches, is incredibly challenging due to profound confounding and scant randomized trial data because of significant ethical issues and logistical barriers. Instead, complex causal inference methods have been used with much work still to be performed (6, 15–17).

A total of 1572 articles were screened with 78 meeting full-text eligibility. The panel did not find any clinical studies directly addressing which of the two approaches—clinician judgment or first come, first served—is the most effective during surge conditions. The absence of direct evidence to address this question stems from the fact that, in real-world clinical practice, clinicians use a combination of time-based and clinical judgment-based inputs, which are difficult to quantify. Consequently, we were unable to make an evidence-based recommendation.

Given the ethical and logistical complexity involved in prioritizing ICU admissions, designing a study to address this PICO question poses significant challenges, as it would require controlled, real-time decision-making across diverse clinical contexts, which may be unattainable and ethically impossible to overcome. However, this question could potentially be explored through mathematical models applied to large, retrospectively or more likely prospectively collected datasets, utilizing pre-set rules to minimize selection bias, and factoring in bed capacity constraints to reflect real-world ICU dynamics more accurately. One simulation study, using operational modeling and queueing theory with data generated from a single-center ICU, attempted to address this problem (18). The investigators considered first come, first served with specific thresholds to admit those with survival advantage, or first come, first served with incremental survival advantage and adjusting dynamically according to bed availability. A 1.2% marginal improvement was reported with the more complex dynamic model. This study suggests that utilizing large triage multistudy databases (19) applying mathematical models assisted by recently introduced artificial intelligence tools may be one solution to this question in the future (18).

PICO 2. Use of Objective Triage Tool

We make no recommendation regarding the use of a formal triage tool vs. no use of a triage tool in situations of limited ICU bed capacity due to insufficient evidence.

Rationale. Application of an objective or formal tool to the process of triaging patients to the ICU has the potential benefit of enhancing consistency and fairness, reducing variation in practice, and minimizing the potential for bias. Proposed frameworks for triage have suggested approaches using eligibility or prioritization criteria to serve as tools for this process (20). Consensus-derived objective criteria related to vital signs and admission criteria have been proposed, although there is no clear single objective tool that has emerged (6). The panel sought to understand whether a formal triage tool should be used vs. no triage tool in situations of limited ICU bed availability for critically ill adult patients.

A total of 572 articles were screened with 82 meeting full-text eligibility. The panel further evaluated eight studies (21–28) that potentially addressed the

application of a formal triage tool. Ultimately, these studies provided no direct evidence to support the PICO question. These studies did not meet inclusion criteria because true surge conditions were not tested, non-ICU patients were included, outcomes for mortality predictiveness rather than triage decisions were tested, and objective tools unrelated to triage or ICU admission were used.

PICO 3a. Boarding of Patients When No ICU Space Available

We make no recommendation regarding the transfer of ICU patients to another facility in the setting of limited ICU bed capacity due to insufficient evidence with very low certainty.

Rationale. Limited ICU bed availability poses a critical challenge for hospitals. During crisis-level shortages, clinicians are often compelled to consider interfacility transfer as an option to secure appropriate care for critically ill patients. However, evidence guiding this decision remains scarce.

The panel screened a total of 620 articles with 51 meeting full-text eligibility and one included as supporting evidence. A nonrandomized study by Duke and Green (29) found no significant difference in mortality between 73 patients who underwent interfacility transfer (18; 24.7%) compared with 73 patients who did not (13; 17.8%; **Fig. S5**, <https://links.lww.com/CCM/H862>). They reported a mean increase in length of ICU stay and delay in ICU admission (mean difference [MD], 15.0 d; 95% CI, 1.36–31.36 d and MD, 1.5 hr; 95% CI, 0.79–2.21 hr), respectively, in patients who underwent interfacility transfer compared with those who did not (**Figs. S6 and S7**, <https://links.lww.com/CCM/H862>). The study (29) also reported a longer length of hospital stay among patients who underwent interfacility transfer (MD, 1.7 d; 95% CI, –0.88 to 4.28 d) compared with patients who did not (**Fig. S8**, <https://links.lww.com/CCM/H862>).

The panel determined that the balance of benefits and risks associated with interfacility transfer in this context remains uncertain (summary of evidence, **Table S3**, <https://links.lww.com/CCM/H862>). Mortality outcomes do not appear to differ between transferred and nontransferred patients, but the potential for increased hospital length of stay introduces cost and resource implications that must be considered. Furthermore, the studies do not account for the quality of care provided

to other patients affected by bed shortages at the originating hospital or the effect of transfer on the receiving facility's resources and care standards. It is not known how the variability of ICU capacity constraints across health systems and the potential influence of health system structure (e.g., public vs. private), patient payor status, and geographical and transportation access to alternative facilities affect access to transfer or the decision itself. Transfer feasibility also varies significantly during short-term disasters, such as severe weather events, or prolonged crises, such as pandemics, highlighting the need for case-by-case decision-making. The panel judged that the resources required to facilitate interfacility transfer are large. These could not only impede transfer but also create unfairness in the process. A study of the interregional transfers during the COVID-19 pandemic coordinated by the French Regional Health Authority demonstrated a significant lower likelihood of transferring patients requiring noninvasive ventilation/high-flow oxygen therapy, prone invasive mechanical ventilation, renal replacement therapy, extracorporeal membrane oxygenation, and even having higher Charlson Comorbidity Index (30). The panel was uncertain about whether the intervention would be acceptable to stakeholders or feasible to implement. Insurance coverage may adversely impact patients with low social economic status (evidence-to-decision worksheet, **Table S4**, <https://links.lww.com/CCM/H862>).

PICO 3b. Designated Non-ICU Areas During Surge Conditions

We make no recommendation regarding the availability of designated non-ICU areas to board ICU patients during surge conditions due to insufficient evidence.

Rationale. Many organizations and health systems are in a constant state of surge, whether from day-to-day normal operations, short-term disasters (e.g., hurricanes) or longer-term disasters as with the COVID-19 pandemic. In the setting of limited ICU availability as defined by the studies, we investigated whether having a designated non-ICU area was better than having no designated non-ICU area for overflow in hospitals with ICUs.

The panel screened a total of 620 articles with 51 meeting full-text eligibility. While 51 studies were assessed for eligibility, none had a study design

that directly compared the outcomes of critically ill patients cared for in a designated non-ICU area vs. no designated non-ICU area. Most studies reviewed were descriptive, lacked a comparator, had no outcomes data, or compared patients boarding in non-specialty ICUs. Despite the clinical importance of our PICO question, none of the studies directly addressed it. As a result, there is insufficient evidence to make a recommendation regarding whether hospitals should have designated or no designated non-ICU areas to board adult critically ill patients in the setting of limited ICU bed capacity created during surge conditions.

PICO 4. Who Should Care for ICU Patients in Non-ICU Areas?

We make no recommendation regarding whether patients boarded in non-ICU areas should be cared for by ICU trained practitioners vs. the usual practitioners for that area due to insufficient evidence.

Rationale. Who should care for critically ill patients during times of surge when patients are boarding outside of an ICU was a question that was often asked during the COVID-19 pandemic. During those times, many organizations used “tiered” staffing models whereby care was led by critical care practitioners (medical doctors, registered nurses, advanced practice providers, respiratory therapists, etc.), but providers and teams were often recruited from areas that were less impacted by COVID-19. Even in nonpandemic times, it is a question that arises during ICU bed shortages or surge conditions on a near daily basis at health systems as patients board in emergency departments or general floors waiting for a bed in the ICU.

The panel screened a total of 620 articles with 51 meeting full-text eligibility. While 51 studies were assessed for eligibility, none of the studies directly addressed the above question. One study by Mitarai et al (31) evaluated the care of critically ill patients in the emergency department awaiting a medical ICU (MICU) bed. However, the study was excluded since the interventional group was cared for by ICU trained emergency medicine specialists while comparison groups were cared for by MICU physicians. Thus, the panel makes no recommendation regarding whether patients boarded in non-ICU areas should be cared for by ICU trained practitioners vs. the usual practitioners for that area.

PICO 5. Involvement of Palliative Care Services During Limited ICU Bed Availability

In critical care patients at high risk of dying, we suggest that palliative care services be involved early as compared with no palliative care services or usual care to help reduce length of stay (conditional recommendation, very low certainty evidence).

Rationale. The panel considered that the question of whether early involvement of palliative care services during time of limited resources was important. It aimed to assess whether integrating palliative care early on could not only enhance the quality of care but also improve resource utilization during periods of strained ICU resources. We explored mortality, quality of patient care at the end of life, family satisfaction, and resource utilization.

A total of 163 studies were screened with 29 meeting full-text eligibility and 11 included as supporting evidence. A meta-analysis of eight studies (32–39) found palliative care involvement compared with no palliative care involvement in critically ill patients may have no effect on the risk of mortality (risk ratio, 1.02; 95% CI, 0.77–1.35) but the evidence is very uncertain (**Fig. S9**, <https://links.lww.com/CCM/H862>). A summary of the types of palliative care services provided in the studies included for analysis are presented in **Table 2**. A summary of findings from the studies is presented in **Table S5** (<https://links.lww.com/CCM/H862>).

A meta-analysis of two studies (41, 42) demonstrated very uncertain effect in the quality of dying and death (QODD) based on the QODD survey (MD, 0.21 points; 95% CI, –3.51 to 3.94 points), in critically ill patients with palliative care involvement compared with no palliative care involvement (**Fig. S10**, <https://links.lww.com/CCM/H862>).

A meta-analysis of seven studies (32–34, 37, 38, 40, 41) demonstrated palliative care involvement compared with no palliative care involvement may reduce the length of hospital stay (MD, –2.24 d; 95% CI, –4.11 to –0.37 d), in critically ill patients but the evidence is very uncertain (**Fig. S11**, <https://links.lww.com/CCM/H862>). Another meta-analysis of ten studies (32–35, 37–42) demonstrated a potential reduction in the length of ICU stay (MD, –1.9 d; 95% CI, –2.32 to –1.48 d), in critically ill patients with palliative care involvement compared with no palliative care involvement but similarly the evidence is very uncertain (**Fig. S12**, <https://links.lww.com/CCM/H862>).

TABLE 2.
Summary of Palliative Care Services Provided in the Studies

Study	PC Involvement
Ahrens et al (32), 2003; Mosenthal et al (36), 2008	Structured communication with families of seriously ill patients by communication team
Campbell and Guzman (40), 2003; Campbell and Guzman (39), 2004	Identification of patient's advance directives or preferences about end-of-life care. Assistance with discussion of the prognosis and treatment options. Implementation of PC strategies when treatment goals change to a focus on comfort measures
Curtis et al (41), 2008; Curtis et al (42), 2011	Clinician education, local champions, academic detailing, feedback to clinicians, and system support. Not targeted to patients or family members
Daly et al (33), 2010; Lilly et al (35), 2003	Intensive communication system for family decision-makers of long-stay ICU patients. Each meeting addressed medical update, values and preferences of the patient, goals of care, treatment plan, and milestones
Hsu-Kim et al (34), 2015	PC consultation with patient
Norton et al (37), 2007	Basic PC consultation consisting of: 1) chart review, 2) history of present illness, 3) discussions with medical ICU team, 4) review of PC recommendations with attending physician, and 5) completion of assessment form with PC recommendations. Complete PC consultation consisting of: 1) basic PC consultation, 2) regular involvement by the PC team with patient's family members, 3) full involvement of PC physician, 4) regular involvement of PC team in the patient's treatment, and 5) availability of PC team for additional support for the patient and family as needed
Ma et al (38), 2019	PC consultation comprising of regular visits by PC team. PC consultation included: 1) chart review, 2) meeting with patient, 3) identifying physical and emotional needs of patient and family, 4) PC plan, 5) communication with all parties regarding goals, values, and treatment decisions, and 6) follow-up with patient until discharge

PC = palliative care.

A meta-analysis of two studies (41, 42) demonstrated little to no difference in family satisfaction with care based on the Family Satisfaction in the ICU survey (MD, 0.85 points; 95% CI, -1.99 to 3.7 points), in families of critically ill patients with palliative care involvement compared with no palliative care involvement but the evidence was very uncertain (**Fig. S13**, <https://links.lww.com/CCM/H862>).

The overall certainty of evidence was very low due to risk of bias, inconsistency, and imprecision. While evidence from included studies was inconsistent and retrospective in design, the recommendation reflects the panel's view that defining and aligning goals of care with patient's values and preferences is vital to a critical illness-related hospitalization. The desirable effects were weighed against the absence of observed harm and the overarching priority of ensuring dignified care at the end of life. As such, the panel judged that the balance between the desirable and undesirable effects probably favor early involvement of

palliative care services. The required resources would vary depending on the model of service used (i.e., some models may be more costly to implement). Given the lack of data, the panel was uncertain about whether early palliative care would be cost-effective to implement. The panel judged that the impact on health equity would probably be increased because those with higher socioeconomic status tend to advocate for themselves or their family members to obtain such services. Paradoxically, the panel also felt that in some areas, lower-income individuals may be more informed around death and dying and goals of care. Some ethnic groups may be resistant to palliative care involvement perhaps due to mistrust in the system because of historical injustices. Last, the panel judged that early palliative care services would probably be acceptable to key stakeholders and feasible to implement in clinical practice (evidence-to-decision worksheet, **Table S6**, <https://links.lww.com/CCM/H862>).

TABLE 3.
ICU Triage Research Priorities

Topic	Research Priorities
Clinician judgment vs. time-based admission	Explore mathematical models' applications in large databases (e.g., operational modeling and queueing theory) Study the use of different types of artificial intelligence in real-time decision-making across diverse clinical contexts (e.g., crisis-level resource shortages)
Use of objective triage tool	Evaluate the impact of the use of objective triage tools in resource-limited situations on overall (system and population level) rates of survival, as well as factors such as consistency of decision-making across providers, equity, and fairness
Boarding of patients when no ICU space available	High-quality studies regarding the transfer of ICU patients to another facility in the setting of limited ICU bed capacity Effects on healthcare systems, patient-centered outcomes, and cost efficiency Broader availability and access to data from existing hospital systems or insurance companies for research
Designated non-ICU areas during surge conditions	High-quality observational studies of current practices. Randomized studies would be preferred but are unlikely Evaluation of timing of care delivery and quality outcomes, including to adherence to guidelines as compared with traditional settings Evaluation of cost and satisfaction (patient and staff)
Who should care for ICU patients in non-ICU areas?	High-quality outcome studies of patients in non-ICU areas when cared for by ICU providers (all team members) vs. non-ICU team members For physicians, a comparison of board certification in critical care vs. not. For other providers/team members like advanced practice providers, registered nurses, etc. An exploration of core competencies
Involvement of palliative care services	Studies better defining resource requirements for the intervention and evaluating cost-effectiveness Randomized trials, cost-effectiveness analyses, and strategies to integrate palliative care seamlessly into ICU workflows to determine the best ways to support critically ill patients and their families during times of limited capacity

RESEARCH AGENDA

Crisis-level shortages significantly disrupt patient care. During such incidents, critically ill patients should have access to essential critical care services. However, despite the conceptual importance of triage in minimizing adverse events and improving outcomes, there is insufficient evidence to guide practice recommendations in the critical clinical scenarios considered by the task force. The absence of evidence to inform how to undertake triage further emphasizes the need for process guidelines and data monitoring with transparent reporting and mechanisms for accountability to foster ethical triage should it be required. Future research is paramount on several of the topics covered by these guidelines, with a summary of research priorities presented in **Table 3**. Until more

advanced clinical decision-support tools are available, the final judgment regarding ICU triage should remain with the treating clinician, guided by the fundamental principles of triage applied to local context and available resources, and aligned with the patient's wishes and values.

ACKNOWLEDGMENTS

The Guidelines in Intensive Care Development and Evaluation (GUIDE) group provided methodological support throughout the guidelines development process. We extend our sincere gratitude to Kusum Matthews, MD, MPH, MSCR, for her invaluable contributions to the development of these clinical practice guidelines. Dr. Matthews' expert insights and willingness to share their clinical expertise and time have been instrumental in

shaping these guidelines. The guidelines leadership would like to acknowledge Society of Critical Care Medicine (SCCM) staff, Hariyali Patel, MHA, and Chandni Zabiegala, RRT, and American College of Critical Care Medicine (ACCM) Board of Regents, Karin Reuter-Rice, PhD, NP, FCCM, and Ana Lia Graciano, MD, FCCM, for project management support throughout the guidelines development process. Additionally, Dan Woznica, PhD, for his work on the implementation toolkit work for these guidelines. The ACCM, which honors individuals for their achievements and contributions to multidisciplinary critical care medicine, is the consultative body of the SCCM, which possesses recognized expertise in the practice of critical care. The ACCM has developed administrative guidelines and clinical practice parameters for the critical care practitioner. New guidelines and practice parameters are continually developed, and current ones are systematically reviewed and revised. Librarian services, systematic review, and analysis for these guidelines were provided contractually through the GUIDE Group, McMaster University, Canada. Methodologists served as expert panel members specializing in this area.

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Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's website (<http://journals.lww.com/ccmjjournal>).

Funding for these guidelines was solely provided by the Society of Critical Care Medicine.

Dr. Anesi received research funding from the National Institutes of Health (K23HL161353, R01HL166269), the University of Pennsylvania Perelman School of Medicine Thomas B. McCabe and Jeannette E. Laws McCabe Fund, and the Leonard Davis Institute of Health Economics. Dr. Gallagher is an advisor for Comind, a speaker for Medtronic, and an advisor for Philips Medical. The remaining authors have disclosed that they do not have any potential conflicts of interest.

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